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TRANSPORT PHENOMENA II, CHE 312

EXAM 1

CLOSED BOOK, TIME: 75 MINUTES

A Newtonian fluid with constant physical properties is in a laminar flow in a circular tube of inner radius R according to the following velocity profile

$$v_z = v_{z,max} \left[1 - \left(\frac{r}{R} \right)^2 \right] \quad N_z(R)$$

The fluid is highly viscous, leading to substantial energy dissipation due to viscous effects.

Answer the following questions in terms of the constant parameters:

$v_{z,max}$: maximum velocity of the fluid

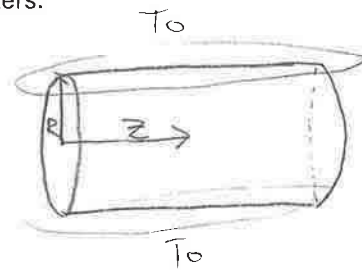
R : tube's inner radius

μ : viscosity of the liquid

ρ : density of the liquid

c_p : isobaric heat capacity per unit mass of the fluid

k : thermal conductivity of the fluid



- 15/20 X1- Using the **appropriate** equation of energy and the dissipation function, write a partial differential equation (PDE) describing the fluid's temperature profile as a function of r and z . $T = T(r, z)$

If the temperature of the tube's inner surface is maintained constant at T_0 , the asymptotic (far enough from the inlet) temperature profile within the fluid becomes independent of z .

- 10/10 X2- Write an ordinary differential equation (ODE) describing the fluid's asymptotic temperature profile as a function of r . $T = T(r)$

- 15/30 X3- Solve the above ODE and apply appropriate boundary conditions to find the fluid's asymptotic temperature profile as a function of r . $T(r) = ?$

- 10/20 X4- Determine the fluid's bulk temperature T_b based on the asymptotic temperature profile derived above.

- 5/20 X5- Calculated the Nusselt number $Nu = \frac{zhR}{k}$ using the heat transfer coefficient based on the asymptotic bulk temperature $q_r|_{r=R} = h(T_b - T_0)$.

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Solve ODE: $\frac{1}{r} \frac{d^2 T}{dr^2} = -\frac{\mu}{k} \left(\frac{dV_z}{dr} \right)^2$

$\frac{1}{r} \int \frac{d^2 T}{dr^2} = -\frac{\mu}{k} \int r \left(\frac{dV_z}{dr} \right)^2$ integrate once $-2\left(\frac{r}{R}\right)^1 \cdot -1$

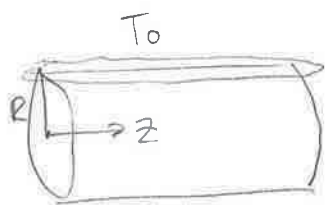
$V_z = V_{z,max} \left[1 - \left(\frac{r}{R} \right)^2 \right]$

$\frac{dT}{dr} = -\frac{\mu}{k} \int r \left(\frac{dV_z}{dr} \right)^2 = -\frac{\mu}{k} \int r \left(V_{z,max} \left[1 - \left(\frac{r}{R} \right)^2 \right] \frac{d}{dr} \right)^2 dr$

$= -\frac{\mu}{k} \int r \left(V_{z,max} \left[2\left(\frac{r}{R}\right) \right]^2 \right) dr$

$= -2 \frac{\mu V_{z,max}}{k} \int r \cdot \frac{r^2}{R^2} dr$

$= -2 \frac{\mu V_{z,max}}{k} \int \frac{r^3}{R^2} dr$



$\frac{dT}{dr} = -2 \frac{\mu V_{z,max}}{k R^2} \cdot \frac{1}{4} r^4 = -\frac{1}{2} \frac{\mu V_{z,max}}{k R^2} r^4 + C_1$

$T(r) = -\frac{1}{2} \frac{\mu V_{z,max}}{k R^2} \cdot \frac{1}{5} r^5 + C_1 r + C_2$

$T(r) = -\frac{1}{10} \frac{\mu V_{z,max}}{k R^2} r^5 + C_1 r + C_2$

B.C $\left\{ \begin{array}{l} T(r=R) = T_0 \rightarrow T_0 = -\frac{1}{10} \frac{\mu V_{z,max}}{k R^2} \cdot R^5 + C_1 R + C_2 \\ \frac{dT}{dr} = 0 \bigg|_{r=0} \quad C_1 = 0 \end{array} \right. \quad C_2 = T_0 + \frac{1}{10} \frac{\mu V_{z,max}}{k R^2} \cdot R^5$

$T(r) = -\frac{1}{10} \frac{\mu V_{z,max}}{k R^2} r^5 + T_0 + \frac{1}{10} \frac{\mu V_{z,max}}{k} R^3$

$\rightarrow T(r) = T_0 + \frac{1}{10} \frac{\mu V_{z,max}}{k} \left(-\frac{r^5}{R^2} + R^3 \right)$

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⑤ $Nu = \frac{2hR}{k}$

$q_r|_{r=R} = h(T_b - T_o)$

$T_b = 2 \frac{T_o}{R^2} + \frac{1}{10} \frac{\mu V_{z,max}}{k} \left(-\frac{1}{3} \frac{r^4}{R} + 2 \frac{R^3}{r^2} \right) \frac{6}{3} - \frac{1}{3} = \frac{5}{3}$

$T_b(r=R) = 2 \frac{T_o}{R^2} + \frac{1}{10} \frac{\mu V_{z,max}}{k} \left(-\frac{1}{3} R^3 + 2R \right)$

$T_b(r=R) = 2 \frac{T_o}{R^2} + \frac{1}{10} \frac{\mu V_{z,max}}{k} \left(\frac{5}{3} R^4 \right)$ solve $T_b(r=R)$

$q_r = h \left(2 \frac{T_o}{R} - \frac{T_o R}{R} + \frac{1}{10} \frac{\mu V_{z,max}}{k} \left(\frac{5}{3} R^4 \right) \right)$ Plug into q_r

$q_r = h \left(2 \frac{T_o - T_o R}{R} + \frac{1}{10} \frac{\mu V_{z,max}}{k} \left(\frac{5}{3} R^4 \right) \right)$ use $q_r = -k \frac{dT}{dr}$

$q_r = -k \frac{dT}{dr} = h(\quad)$

$h = -k \frac{dT}{dr} \left(\frac{1}{R} \right)$

$\frac{dT}{dr} = -\frac{1}{2} \frac{\mu V_{z,max}}{k R^2} r^4 + 0$ solve Nu

$Nu = 2 \left(\frac{\mu V_{z,max}}{k R^2} \right) \left(2 \frac{T_o - T_o R}{R} + \frac{1}{6} \frac{\mu V_{z,max} R^4}{k} \right) R$

$Nu = 2 \left(\frac{\mu V_{z,max}}{R} \right) \left((T_o - T_o R) k + \frac{1}{6} \mu V_{z,max} R^5 \right)$